Real-Time IPC Synchronization Dashboard

Memory leaks in Android applications can cause excessive memory consumption, slow application performance, and unexpected crashes, negatively affecting user experience. This project extends LeakCanary, an existing open-source Android library, to improve memory leak detection and analysis. The proposed system focuses on two features: automatic detection of objects that remain in memory after their lifecycle ends, and leak trace analysis to identify the reference chains preventing their garbage collection. These features will help developers locate the root causes of memory leaks instead of relying solely on memory usage statistics. The system will be evaluated using controlled memory leak scenarios by measuring detection effectiveness and verifying whether the identified reference chains lead to the actual causes. The project aims to simplify memory leak debugging, reduce unnecessary memory consumption, and improve the reliability and performance of Android applications.